$$g_{AA}(r) = \frac{1}{\rho\chi_A}\frac{n_{AA}(r)}{4\pi r^2 dr} = \frac{n_{AA}(r)}{n_{id,A}(r)}$$

$$g_{BB}(r) = \frac{1}{\rho\chi_B}\frac{n_{BB}(r)}{4\pi r^2 dr} = \frac{n_{BB}(r)}{n_{id,B}(r)}$$

$$g_{AB}(r) = \frac{1}{\rho}\frac{n_{AB}(r)}{4\pi r^2 dr} = \frac{n_{AB}(r)}{n_{id}(r)}$$